

CLASS 8 SCIENCE — REVISION NOTES

Chapter 2: Microorganisms & The Cell | NCERT Curriculum

Reading the relevant NCERT chapter before using these notes is highly recommended. This chapter is about tiny organisms and how they affect us.

1. LENSES & MICROSCOPES

LENS : Lens refer to a piece of glass that is shaped like a lentil seed, thick at middle & the thinner from sides (This definition is based on this chapter & this structure is of a convex lens). Lenses are used for making microscopes & are very important for research. Early microscopes were not that powerful but led to the discovery of organisms that an unaided eye cannot see.

2. THE CELL & ITS IMPORTANCE

CELL : Cell is the smallest structural & functional unit of life. Every living being is made up of cell(s). Some organisms are made up of one cell only and are called **unicellular organisms** & some organisms have more than one cell, they are known as **multicellular organisms**.

WHY CELL IS THE BASIC UNIT OF LIFE? : Cell is the basic unit of life because cells can work independently & control their growth & energy by themselves. This is also the reason why unicellular organisms survive, because they can do every function by themselves.

3. THREE MAIN PARTS OF THE CELL

- **CELL MEMBRANE** : It separates a cell from another cell. It is also porous so it allows entry of substances & exit of waste.
- **CYTOPLASM** : It is present inside the cell membrane and is a jelly-like liquid which has all the essential nutrients. Most of the life processes happen in it.
- **NUCLEUS** : It regulates all activities inside the cell like hormonal activities and growth. Bacteria don't have it, instead they have Nucleoid.

4. SPECIAL PARTS OF THE CELL

- **CELL WALL** : It is present in plant and fungi cells. It acts as a covering for the whole cell. It provides rigidity and support for standing upright. It is also porous.
- **PLASTIDS** : Tiny rod-like structures in plant cells. Eg – Chloroplasts. In non-green plants, they act as small storage boxes.
- **VACUOLE** : It looks like an empty space in cell but it is very important for plant cells as it helps them maintain their shape and stay in arranged patterns like seen in onion cells. It is rarely present in animal cells. But when it is, it is very small. In plants, it takes up about 90% of cell's space.

5. COMPARATIVE TABLES

PLANT CELLS VS ANIMAL CELLS

PLANT CELLS	ANIMAL CELLS
They have cell walls.	They don't have cell walls.
They have plastids.	They don't have plastids.
They have one vacuole that covers 90% of the cell's space.	Vacuoles are small and there can be many.

MUSCLE CELL VS NEURON (NERVE CELL)

MUSCLE CELL	NEURON (NERVE CELL)
Spindle shaped appearance.	Long and branched appearance.
Thin and flexible, works by contraction and relaxation.	Transports messages and signals from brain to other parts or vice versa.
Eg – Heart, Digestive system	Eg – Neurological system

6. LEVELS OF ORGANIZATION & EGG CELL

- **LEVELS** : Cells ⇒ Tissue ⇒ Organ ⇒ Organ System ⇒ Organisms
- **EGG** : It is a cell that can repeatedly divide to form a complete living organism that is made up of many cells. Largest cell ever found is the yolk of an ostrich egg (ranging from 150-170 mm).

7. MICROBES (MICROORGANISMS)

Some organisms that cannot be seen with unaided eyes are called micro-organisms or microbes. Eg – Fungi. They are usually observed using a microscope. Even air, water, soil suspension and stagnant water, all have microbes.

MICROBES COMMONLY FOUND IN :

- **STAGNANT WATER** : Amoeba (Protozoa), Paramecium (Protozoa), Algae
- **SOIL SUSPENSION** : Mould (Fungi), Bread Mould (Fungi), Algae, Bacteria

TYPES OF MICROBES :

1. Fungi
2. Protozoa
3. Virus
4. Bacteria
5. Algae

USES OF MICROBES :

- Used in Baking and Alcohol Industries (Yeast is the name of that microbe)
- Lactobacillus is commonly used to curdle the milk.
- Microalgae produce more than 50% of Earth's oxygen. They are also used to make health supplements. Eg – Spirulina as a source of protein and Vitamin B12.

- Rhizobium is a bacterium which lives in the root nodules of legumes and does nitrogen fixation (converting atmospheric nitrogen into nitrogenous compounds). This ultimately increases the nitrogen in the soil and helps the growth of crops.

HARMFUL EFFECTS OF MICROBES :

- They can cause diseases that can also be life threatening. Eg – Malaria
- They also spoil things. Eg – Mould on bread. That is why preservation methods like pickling are used (adding a lot of salt or sugar or spices, dehydrates the substance, thus protecting from microbial growth).

8. KEY SCIENTISTS & DISCOVERIES

- **ROBERT HOOKE** : A scientist and a part of the ROYAL SOCIETY. Published a book called *Micrographia*, in which he used the word 'cell' for describing the basic unit of life. He used a microscope to see beyond our eyes and drew detailed diagrams of what he observed.
- **ANTONIE VAN LEEUWENHOEK** : A Dutch scientist known as the "Father of Microbiology" as he made more powerful microscopes and saw things like blood cells and bacteria and described his sighting.
- **ANAND MOHAN CHAKRABARTY** : He was an Indian scientist who got a patent for developing a special bacterium that can help break down spills and help clean the environment.